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Synopsis
Smart Water Irrigation System

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Bachelor of Technology
In
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by

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This is to certify that **Vidula Patil, Purvaja Chaudhari, Pratham Sankhe, Harsh Kini** are bonafide students of the Bachelor of Engineering programme in Computer Science and Engineering (Data Science), Department of Computer Science and Engineering (Data Science), **St. John College of Engineering and Management**, University of Mumbai. They have prepared the synopsis titled "**Smart Water Irrigation System**" as part of the academic requirements, subject to review and approval by the institute.

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Declaration

we declare that this written submission represents our ideas in my/our own words and where others ideas or words have been included, we have adequately cited and referenced the original sources. we also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. we understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

The Smart Irrigation System is a compact hardware project designed to water a plant automatically according to soil moisture. The system uses a soil moisture sensor module, a relay module, a 5V water pump, a 4×AA battery holder, connecting wires and a water pipe. When the soil becomes dry, the sensor output causes the relay to switch the pump on, delivering water to the pot. As moisture rises to the sensor's switching threshold, the pump is expected to turn off. The project demonstrates a simple feedback-based automation concept for home gardening, nurseries and educational use. The supplied presentation describes a direct-connect, continuously active arrangement. Exact wiring, voltage compatibility, pump current, calibration and observed performance must be verified on the physical prototype before final submission. This synopsis presents the design intent and planned validation; it does not claim measured water savings or reliability results.

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ABBREVIATIONS AND SYMBOLS

- DC — Direct Current
- BOM — Bill of Materials
- LED — Light Emitting Diode
- NO — Normally Open
- NC — Normally Closed
- PCB — Printed Circuit Board
- SDG — Sustainable Development Goal
- VCC — Positive Supply Voltage
- GND — Ground
- WBS — Work Breakdown Structure
- IoT — Internet of Things
- V — Volt

CHAPTER 1: INTRODUCTION

1.1 Background

Plants require suitable soil moisture for healthy growth. Manual watering depends on a person remembering when and how much to water. In home gardens and small nurseries, this may result in overwatering or underwatering. A low-cost automatic irrigation circuit can respond to soil conditions and reduce the need for constant manual checking.

1.2 Problem Statement

Manual plant watering may be inconsistent, particularly when the owner is away or caring for several pots. Water may be supplied when the soil is already wet, or missed when it becomes dry. The project addresses this issue through moisture sensing and automatic pump switching. Its performance depends on sensor placement, soil type, threshold adjustment and safe power design.

1.3 Need of the Project

The project provides a practical demonstration of sensor-based control using accessible hardware. It links measurement of soil condition to a switching action and water delivery, making it useful for learning electronics, automation and basic control systems.

1.4 Objectives

The main objective is to develop a compact soil-moisture-controlled irrigation prototype. Specific objectives are:

- Detect soil moisture using the sensor module.
- Activate the water pump when soil moisture falls below the selected threshold.
- Stop watering when the sensor reaches its switching threshold.
- Use readily available, low-voltage components.
- Evaluate response, repeatability, leakage, and safe operation through planned tests.

1.5 Scope

The proposed scope is a single-pot demonstration using a soil moisture sensor module, relay module, 5V water pump, 4×AA battery holder, jumper wires and transparent pipe. It is intended for small plants and educational demonstrations. The current design is not an agricultural-scale irrigation controller and has no confirmed IoT monitoring, data logging or app connectivity.

1.6 Expected Outcomes

The expected outcome is a working prototype that detects dry soil and operates a pump automatically. Testing will document sensor switching behavior, pump activation/deactivation, water delivery, battery behavior and practical limitations. Water conservation is a design goal, not a measured result until comparative trials are conducted.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The project is informed by established concepts in soil-moisture sensing, threshold control, relay switching, small DC pumping and water-efficient irrigation. A full academic survey should compare specific sensor calibration studies and automated irrigation implementations, with verified bibliographic details.

2.2 Literature Survey Table

Preliminary source themes for the literature survey are listed below. Before final submission, replace or supplement these topic summaries with 10–15 verified papers or authoritative sources, recording author, title, year, method and findings.

Topic / source area	Method or focus	Relevance to this project	What should be verified
Soil-moisture sensing and calibration	How probe output changes with soil moisture and placement	Supports selecting and calibrating the moisture threshold	Sensor type, soil dependence, repeatability and calibration method
Automatic irrigation control	Threshold-based switching of a pump or valve	Provides a basis for the dry-soil / watering / stop sequence	Control logic, threshold behavior and watering duration
Resistive versus capacitive probes	Comparison of sensing principles and durability	Helps explain limitations of the pictured probe	Corrosion, response drift and suitability for long-term use
Relay switching of DC loads	Relay contacts used to switch a pump load	Relevant to safe pump activation from a sensor module	Input voltage, contact rating, pump startup current and protection
Low-voltage DC pump systems	Pump voltage, current, flow and priming requirements	Helps check compatibility between the 5V pump and power source	Manufacturer ratings and loaded battery voltage
Water-efficient irrigation practices	Applying water according to plant/soil need	Provides context for the potential water-use benefit	Do not claim savings without comparative measurements
Small-scale automated plant watering	Prototype design and evaluation methods	Provides examples of test cases and system limitations	Repeatable tests, placement, soil type and failure handling

2.3 Research Gap

Low-cost threshold-controlled kits are accessible but may be affected by soil composition, sensor corrosion, inconsistent probe placement, relay chatter, pump current requirements and uncontrolled watering duration. The prototype should therefore validate its switching behavior and safety rather than assume universal moisture accuracy.

2.4 Novelty / Innovation

The project contribution is a practical integration and demonstration of moisture sensing, relay switching and pump-based watering with simple components. It is an educational automation prototype, not a claim of a new sensor or irrigation algorithm.

CHAPTER 3: PROPOSED METHODOLOGY

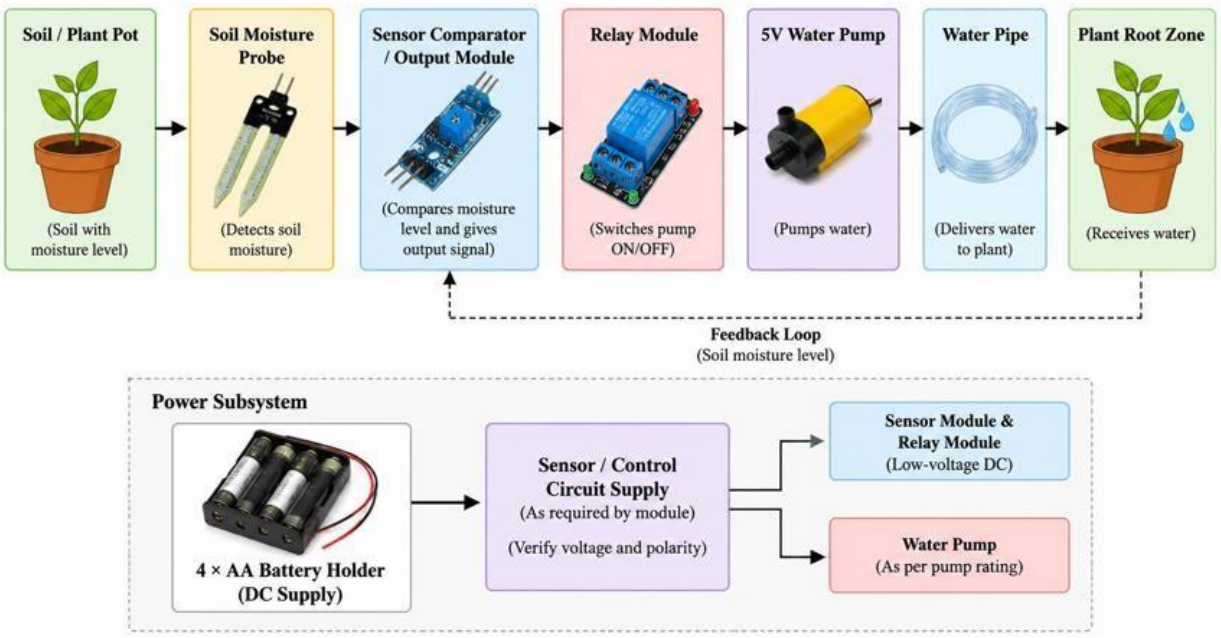
3.1 Proposed System

The system consists of a soil moisture probe and sensor interface, a relay module, a low-voltage water pump, a water source and delivery pipe, and a battery supply. The sensor module monitors soil conditions and provides a control output. The relay acts as an electrically controlled switch for the pump circuit. The supplied presentation lists a 5V pump and 4×AA battery holder; actual supply compatibility and relay contact arrangement must be checked against component specifications before wiring.

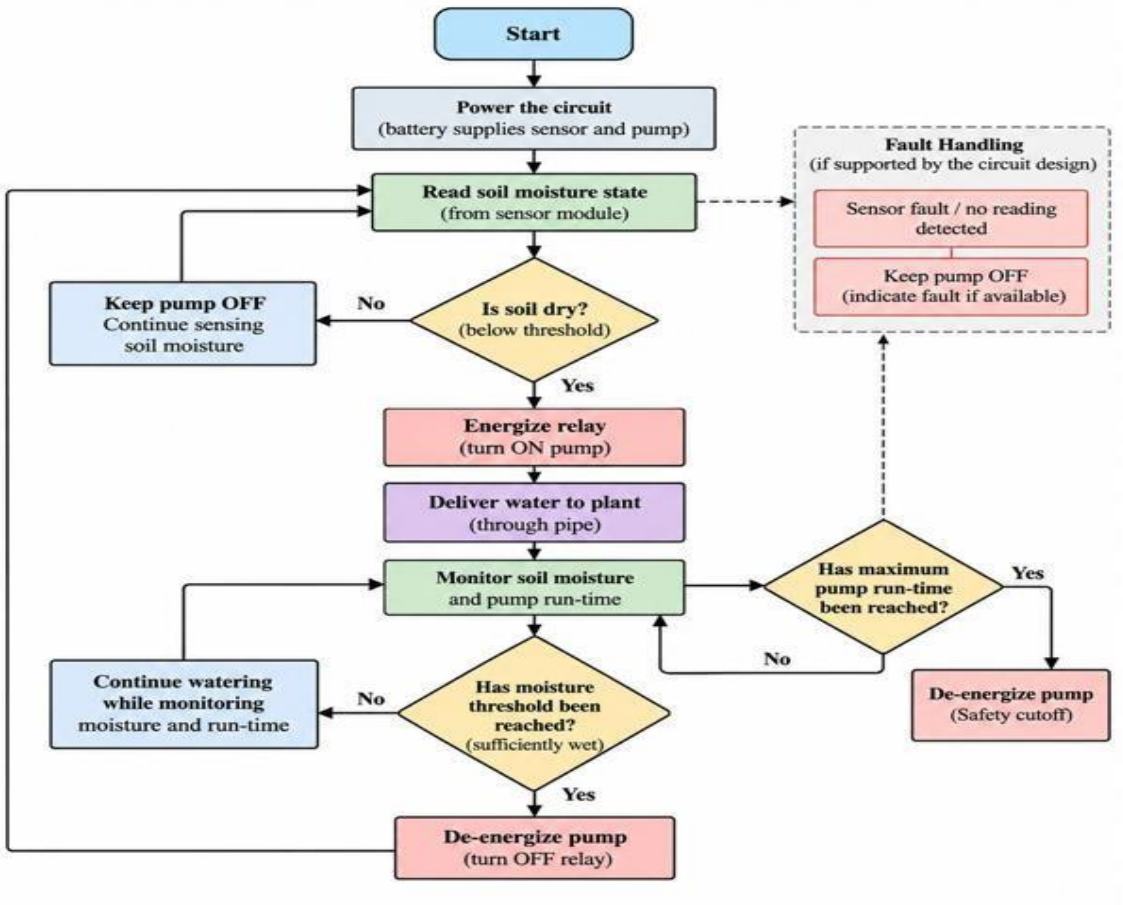
3.2 Working Principle

- Power-up: The battery supplies power to the sensor/control circuit and pump power path as designed.
- Sensing: The probe detects changes in soil moisture.
- Decision: When the soil is dry relative to the module threshold, its output changes state.
- Switching: The relay changes contact state and energizes the pump.
- Water Delivery: The pump moves water through the pipe into the pot.
- Stop Condition: As soil moisture rises past the return threshold, the sensor output changes and the pump switches off. Confirm the actual trigger polarity and threshold behavior experimentally.

3.3 System Architecture / Block Diagram



3.4 Flowchart



3.5 Methodology

The project will be developed and validated in iterative stages:

Stage 1 – Component Inspection: Identify the components and record their model numbers, voltage and current ratings, and terminal labels.

Stage 2 – Functional Wiring: Assemble the soil moisture sensor and relay control circuit, keeping the power supply disconnected during wiring and assembly.

Stage 3 – Dry and Wet Testing: Observe the sensor output under dry and moist soil conditions, perform controlled wetting tests, and adjust the moisture threshold as required.

Stage 4 – Pump Integration: Integrate the water pump using a suitable power supply and safe switching arrangement. Verify the pump's operation and relay response.

Stage 5 – Full Prototype Assembly: Install the water pipe and assemble the complete prototype. Verify that water reaches the plant pot properly while keeping the electronic components protected from moisture.

Stage 6 – Evaluation and Validation: Conduct repeated trials to record pump activation and shutoff behavior, inspect for water leakage, check battery condition, and document the system's performance, limitations, and observations.

3.6 Components / Materials

The proposed hardware, as identified in the supplied presentation and component images, includes: soil moisture sensor probe with comparator/interface module; 1-channel relay module ,5V DC water pump; 4×AA battery holder and suitable batteries; female-to-female jumper wires; 6 mm water pipe; water reservoir and plant pot. Confirm every component's actual rating and compatibility before assembly.

Component / resource	Specification / purpose	Quantity
Soil moisture sensor module	Probe with comparator/interface board; exact model to be recorded	1
Relay module	Single-channel relay board; pictured relay marked SRD-05VDC-SL-C; confirm board input ratings	1
Water pump	5V DC mini water pump, as shown in the project presentation	1
Battery holder and batteries	4×AA holder; verify actual loaded voltage/current suitability	1 set
Jumper wires	Female-to-female jumper wires as shown	As required
Water pipe	6 mm transparent water pipe as shown	As required
Water container and plant pot	Water source and pot for demonstration	1 each
Basic tools and test materials	Multimeter, suitable connectors and dry/wet soil test setup	As available

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure (WBS).

WBS	Deliverable / activity
1	Define project objective and operating conditions
2	Review soil-moisture sensing and automatic irrigation concepts
3	Inspect component labels and verify voltage/current ratings
4	Prepare functional block diagram and verified circuit plan
5	Assemble and calibrate the soil moisture sensor and relay path
6	Integrate pump, water source and delivery pipe
7	Test dry/wet response, pump switching, leakage and power behavior
8	Document observations, limitations, diagrams and final report

4.2 Gantt Chart / Semester Plan

Indicative Project Timeline

The following is the indicative project timeline, which may be adjusted according to the actual academic calendar and work completed.

Weeks 1–2: Requirements definition and background study.

Weeks 3–4: Component verification and circuit planning.

Weeks 5–6: Sensor and relay assembly, followed by threshold adjustment.

Weeks 7–8: Pump and water pipe integration.

Weeks 9–10: Testing, repeated observations, and necessary improvements.

Weeks 11–12: Report preparation, diagram finalization, and presentation.

Activity	W1–2	W3–4	W5–6	W7–8	W9–10	W11–12
Requirements / literature	■					
Component inspection / circuit plan	■	■				
Sensor and relay assembly		■	■			
Pump / pipe integration				■		
Calibration and testing				■	■	
Troubleshooting / repeat tests					■	■
Documentation / presentation						■

4.3 Roles and Responsibilities

Team Members and Responsibilities

: Project coordination, requirement definition, and report compilation.
: Component research, verification of ratings, and circuit documentation.
: Prototype assembly, sensor setup, and water pump integration.
: Testing, observation recording, photography, and presentation preparation.

4.4 Risk Assessment

Risk	Impact	Mitigation
Water near electronics	Short circuit or component damage	Keep electronics away from the reservoir and splashes; disconnect power before adjustments
Incorrect voltage or polarity	Component damage or unsafe heating	Check labels and datasheets; measure supply before connecting
Pump startup current exceeds supply capability	Pump fails to start or battery voltage drops	Verify pump current and battery performance under load
Relay input/contact mismatch	Relay may not switch or contacts may fail	Verify relay board input requirements and contact ratings
Sensor threshold varies by soil / placement	Overwatering or underwatering	Calibrate in the actual soil and fix probe depth/position
Pump continues running / sensor fails	Excess watering or empty reservoir	Supervise tests and add a suitable maximum run-time safeguard if redesigning
Leaks or loose tubing	Water spill and unreliable delivery	Secure tubing and test with low water volume before operation

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The prototype is technically feasible using readily available low-voltage modules. Integration requires confirmation that the sensor output can drive the relay input, that the relay contact rating is suitable for the pump, and that the selected power source meets pump startup and operating current. A typical 4×AA pack voltage varies with battery chemistry and charge state; do not assume it directly provides regulated 5V. Verify the pump and board ratings and provide suitable regulation or a separate supply if required.

5.2 Economic Feasibility

The project is intended to be low-cost and uses common hobby components. Final cost depends on locally purchased modules, batteries, tubing, wiring, enclosure and any added protection or regulation. Record actual receipts or quotations rather than using assumed market prices.

5.3 Operational Feasibility

Once calibrated and correctly assembled, the system is intended to operate automatically with limited user intervention. The user must refill the water source, inspect tubing, monitor battery condition and periodically check sensor performance. Safe placement and protection from splashes are essential.

5.4 Budget Estimation

Item	Quantity	Actual cost (₹)
Soil moisture sensor module	1	
Relay module	1	
5V DC water pump	1	
4×AA battery holder and batteries	1 set	
Female-to-female jumper wires	As required	
6 mm water pipe	As required	
Water container / pot / misc. fittings	As required	
Total	—	

5.5 Bill of Materials (BOM)

Resource	Specification / description	Quantity
Soil moisture sensor	Probe and sensor interface/comparator board; confirm exact model	1
Relay module	Single-channel relay module; verify board input and contact ratings	1
Water pump	5V DC mini pump shown in the presentation	1
Battery holder	4×AA battery holder; supply suitability must be checked	1
Batteries	Four AA cells compatible with the holder	4
F-F jumper wires	Female-to-female leads shown in the component image	As required

Water pipe	6 mm transparent pipe shown in the component image	As required
Water source / plant pot	Container for water and pot for demonstration	1 each

CHAPTER 6: EXPECTED RESULTS & IMPACT

6.1 Expected Outcomes

The expected outcomes are:

- Sensor output changes consistently between dry and adequately moist soil conditions after calibration.
- Relay and pump switch in response to the sensor output.
- Water is delivered through the pipe to the target pot.
- Pump stops at the selected moisture threshold, subject to sensor response and placement.
- Trial records document response, repeatability, leakage, and power behavior.

These outcomes must be confirmed through actual testing.

6.2 Industrial Relevance

The concept is relevant to home gardening, nurseries, greenhouse demonstrations and introductory precision-agriculture education. More advanced commercial irrigation systems require robust sensing, field calibration, weather/soil context, fault reporting and protected outdoor electronics.

6.3 Societal and Environmental Impact

Automated watering may help users maintain plants and avoid unnecessary watering when calibrated appropriately. Environmental benefit is possible but should not be quantified without comparative measurements. Improper thresholds or leaks can waste water, so responsible testing and operation are necessary.

6.4 Sustainability and SDG Mapping

Potential alignment, subject to institutional framing:

- SDG 6 — Clean Water and Sanitation: introduces water-conscious irrigation practices.
- SDG 9 — Industry, Innovation and Infrastructure: demonstrates accessible automation.
- SDG 12 — Responsible Consumption and Production: encourages mindful water use.

Alignment is aspirational; no measured SDG impact is claimed.

6.5 Future Scope

Future improvements may include a capacitive moisture sensor, microcontroller-based control, hysteresis and timed safety cutoff, solar power with proper charge management, multiple-zone irrigation, data logging, IoT notifications, weather-aware scheduling and rugged enclosure. Any upgrade should be evaluated for cost, power consumption and reliability.

CHAPTER 7: CONCLUSION

7.1 Conclusion

The Smart Irrigation System proposes a simple automatic plant-watering solution by combining a soil moisture sensor, relay module and DC water pump. It demonstrates a feedback-driven response to soil condition and can serve as an educational prototype for automation and water-use awareness. Completion quality depends on verified wiring, compatible power, calibration and documented testing.

7.2 Limitations

The present concept is designed for a small pot, not large-scale farming. Resistive probes can corrode and sensor readings vary with soil type and placement. A threshold module may provide only coarse control. Pump current, battery suitability, relay input compatibility and exact wiring must be verified. No quantified water saving, long-duration reliability or field performance is claimed without test evidence.

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[4] Manufacturer datasheet and user documentation of the soil moisture sensor module used in the prototype. (Add the exact manufacturer, model number, and document details after verification.)

[5] Manufacturer datasheet and specifications of the DC water pump used in the prototype. (Add the exact manufacturer, model number, and document details after verification.)

APPENDICES

Appendix A: Proposed Component and Connection Details

The following table lists the major components of the Smart Irrigation System, their connection points, proposed connections, and important remarks for prototype assembly.

Component / Module	Device Pin / Terminal	Proposed Connection	Remarks
Soil Moisture Sensor Module	VCC / GND	3.3V or 5V supply / Common GND	Verify module operating voltage and polarity.
Soil Moisture Sensor Module	OUT / Signal	Relay control input or suitable interface	Confirm output type, trigger polarity, and voltage compatibility.
Relay Module	VCC / GND	Compatible control supply / Common GND	Verify relay module voltage and current requirements.
Relay Module	IN	Sensor output or suitable driver	Ensure compatibility between sensor output and relay input.
Relay Module	COM / NO	Pump power switching path, as designed	Select contact arrangement according to the required default state and circuit design.
Relay Module	NC	Not used in the proposed arrangement	Use only if required by the verified circuit design.
DC Water Pump	Positive (+) / Negative (–)	Appropriately rated pump supply and switched path	Verify rated voltage, startup current, polarity, and operating current.
Battery Holder (4 × AA)	Positive (+) / Negative (–)	Suitable power supply arrangement	Verify loaded voltage and whether regulation is required.
Water Pipe	Inlet / Outlet	Pump outlet to plant pot	Check pipe fitment, water flow, and leakage.
Common Ground	GND connections	Shared reference where required by circuit design	Confirm grounding requirements and avoid unintended power connections.